Bioresource Technology 379 (2023) 129040

Contents lists available at ScienceDirect

Bioresource Technology

journal homepage: www.elsevier.com/locate/biortech

Machine learning prediction of contents of oxygenated components in bio-oil using extreme gradient boosting method under different pyrolysis conditions

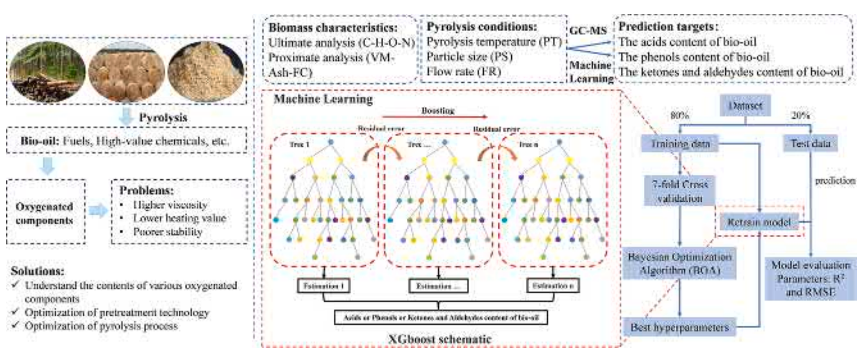
Sheng Su, Juan Wang *

National Key Laboratory of Science and Technology on Aero-Engine Aero-Thermodynamics, School of Energy and Power Engineering, Beihang University, Beijing 100191, China

HIGHLIGHTS

- Predicted contents of components in bio-oil using machine learning.
- Good prediction performance of the XGB algorithm.
- Identified the importance and sensitivity of features.
- PDA analysis provides insight into the pyrolysis characteristics.

GRAPHICAL ABSTRACT



ARTICLE INFO

Keywords:
Oxygenated components
Bio-oil
Machine learning
Biomass characteristics
Pyrolysis conditions

ABSTRACT

This work aims to develop a prediction model for the contents of oxygenated components in bio-oil based on machine learning according to different pyrolysis conditions and biomass characteristics. The prediction model was constructed using the extreme gradient boosting (XGB) method, and the prediction accuracy was evaluated using the test dataset. The partial dependence analysis (PDA) method was used to derive the pattern of influence of each input feature individually or in combination on the output variable. The results show that the prediction models constructed from biomass ultimate analysis and pyrolysis conditions can predict the contents of oxygenated components in bio-oil more accurately than the models constructed from biomass proximate analysis. Moderate C and O contents, higher H content of biomass, lower flow rate, and higher pyrolysis temperature can improve bio-oil quality.

1. Introduction

Achieving net-zero emissions has become a common target to address the global energy crisis and pollution. Even if governments completely adhere to the Announced Pledges Scenario (APS), global CO_2 emissions would be 21 billion tons in 2050, which is still 20 years behind

* Corresponding author.
E-mail address: juanwang@buaa.edu.cn (J. Wang).

https://doi.org/10.1016/j.biortech.2023.129040
Received 23 February 2023; Received in revised form 5 April 2023; Accepted 7 April 2023
Available online 8 April 2023
0960-8524/

the net zero emissions criterion, according to the statistics and predictions from the World Energy Agency (International Energy Agency, 2021). The report also emphasizes that the development of renewable energy is the most efficient approach to closing the gap between the carbon emissions of the two scenarios, the Stated Policies Scenario (STEPS) and APS. Due to its plentiful reserves, environmental friendliness, and renewable nature, biomass has attracted many interest (Vuppaladadiyam et al., 2022). Lignocellulosic biomass, which makes up more than 80% of all biomass, is the most prevalent on earth. Wood (softwood and hardwood), agricultural waste (straw, bran, chaff, etc.), and waste from the processing of forest products (sawdust, etc.) make up the majority of lignocellulosic biomass (Qiu et al., 2022). This kind of biomass feedstock is less expensive than the first-generation biomass energy, which is created from food crops.

Compared with biochemical conversion, thermochemical conversion has the advantages of simple equipment, high bio-oil yield, cheap cost, and quick processing for converting large amounts of biomass (Tripathi et al., 2016). Therefore, this method has attracted much attention in the field of biomass resource utilization. Pyrolysis technology refers to the decomposition of biomass into biochar, non-condensable gases, and bio-oil under anoxic or anaerobic conditions (Dhyani and Bhaskar, 2018). With the advantages of CO_2-neutrality and low sulfur content, bio-oil is regarded as an environmentally friendly alternative fuel that can help alleviate the environmental problems caused by fossil fuel combustion (Rezaei et al., 2014). However, many chemical reactions, including depolymerization, C-C bond breaking, and dehydration, occur during the pyrolysis of biomass, so there are many oxygenated components produced, leading to the higher viscosity, lower heating value, and poorer stability of bio-oil compared with traditional petroleum fuels (Wang et al., 2013a). Therefore, it is important to understand the contents of various oxygenated components for bio-oil upgrading and modification, such as designing specific catalysts to promote bio-oil hydrodeoxygenation and optimizing pretreatment conditions to regulate bio-oil components.

Machine learning algorithms, in particular, have drawn much attention recently for their great predicting powers based on learning from historical data (Ge et al., 2021). It can be used to develop complicated nonlinear relationships between inputs and target outputs more quickly than conventional reaction dynamics models (Sun et al., 2016b). Since 2018, many scholars have tried to apply machine learning to the prediction of bio-oil yield and some macroscopic properties and have achieved some interesting results. The support vector machine (SVM) method's generalization was superior to the artificial neural network (ANN) on the prediction of bio-oil high heating value and yield (Chen et al., 2018). For the hydrogen content prediction in bio-oil, the method of random forest (RF) algorithm was more feasible compared with the multiple linear regression (MLR) (Tang et al., 2020). Similar to that, Xu et al. (2021) achieved this prediction goal using the SVM model enhanced by the Gray Wolf algorithm. Yang et al. (2022) successfully predicted oxygen content in bio-oil using the RF method with biomass properties and reaction conditions as input parameters and found that biomass properties were more important for the oxygen content prediction in bio-oil than reaction conditions. A more thorough study was performed using the RF algorithm by Zhang et al. (2022a), which predicted the elemental content, yield, heating value, and viscosity of bio-oil. Even though many studies have constructed models based on machine learning algorithms to predict bio-oil yields and macroscopic characteristics, machine learning prediction models for detailed information of bio-oil components (e.g., contents of acids, phenolics, and ketones) have been rarely reported. This research is a new attempt to apply machine learning algorithms to predict the contents of components in bio-oil.

Boosting is an integrated learning method in which the training process takes the form of a ladder. The learners are trained one by one in order. The training dataset of each individual learner is transformed according to a certain strategy (e.g., the errors of all previous individual learners are transformed into the training dataset of the next individual learner) (Chen et al., 2016). Thus, the performance of the integrated learner can be improved in this way. The XGB algorithm showed satisfactory performance in bio-oil yield and macroscopic properties prediction. Dong et al. (2023) compared the prediction accuracy of different algorithms for bio-oil yield. The R^2 of the regression models of the RF and XGB algorithms were 0.85 and 0.84, respectively, which were higher than the Adaptive boosting (AdaBoost) algorithm (0.56) and the Gradient Boosting Decision Tree (GBDT) algorithm (0.80). Mathur et al. (2022) predicted bio-oil yields using biomass characteristics and pyrolysis conditions as input features. The prediction model developed using the XGB algorithm resulted in a larger R^2 (0.89) and smaller RMSE (2.39) compared to the RF algorithm. In addition, the XGB algorithm achieved excellent accuracy and successfully avoided under- and over-fitting in the prediction of biochar yield and elemental composition (Pathy et al., 2020). Thus, it seems feasible to use the XGB algorithm to construct a prediction model for the contents of multiple components of bio-oil. Experimental studies on the contents of components in bio-oil usually require the use of instruments such as chromatography and mass spectrometry. This is not only demanding on the equipment, but also expensive. Compared with experimental methods, machine learning prediction models are not subject to these limitations, which can serve as a guide for optimizing the biomass pyrolysis technology.

In this research, the prediction models were constructed for the content of phenols, the content of acids and the total content of ketones and aldehydes in bio-oil using pyrolysis conditions, ultimate analysis and proximate analysis as input features. The first objective is to choose various input features to construct several models, train the models, and compare their performance to improve the prediction models with higher accuracy and resilience. The second objective is to examine the pattern of how each input parameter alone or in combination affects the contents of oxygenated bio-oil components. The XGB method, which is used in this study, is described in detail by Chen et al. (2015b) and Nielsen (2016).

2. Methods

2.1. Dataset creation

The dataset for the contents of acids (118 dataset), the contents of phenols (122 dataset), and the total contents of ketones and aldehydes (112 dataset) used to train and evaluate the model are collected from the papers of the previous studies (Bi et al., 2020; Cen et al., 2019; Chen et al., 2015a; Chen et al., 2017; Chen et al., 2020; Chen et al., 2021; Lu et al., 2020a; Lu et al., 2020b; Setkit et al., 2021; Sun et al., 2019; Tu et al., 2021; Zeng et al., 2019; Zhang et al., 2016; Zhang and Xiong, 2016; Zhang et al., 2019; Zhang et al., 2022b; Zhuang et al., 2022). The data are extracted from the graph using Plot Digitizer or collected directly from the table in the references. To avoid the influence of different experimental equipment, all experimental data are collected from fixed-bed reactors, and the carrier gas is nitrogen. Each dataset includes biomass characteristics, pyrolysis conditions, and contents of oxygenated components in bio-oil. Biomass proximate analysis (fixed carbon content [FC], ash content [Ash], and volatile matter content [VM]) and ultimate analysis (C, N, O, and H) are chosen to represent biomass characteristics. Pyrolysis conditions include pyrolysis temperature (PT), particle size (PS) of biomass and flow rate (FR) of nitrogen. Pyrolysis time and heating rate are also important pyrolysis conditions in biomass pyrolysis reaction. However, the completion of the biomass pyrolysis is indicated by the fact that the gas product no longer increases in most studies, and the pyrolysis time is not recorded. Moreover, the flow rate and the residence time of pyrolysis products are closely related. Therefore, the FR is used instead of the pyrolysis time. In addition, the heating rate is removed to maintain the consistency of the input variables because a portion of the data collected come from thermostatic pyrolysis.

A common method for examining the linear correlation between any two features in a dataset is the Pearson Correlation Coefficient (PCC). Because PCC obeys a two-tailed t-distribution with N-2 degrees of freedom, the significance level can be quantified by Eq. (1).

$$t = \frac{r\sqrt{N\text{-}2}}{\sqrt{1\text{-}r^2}} \tag{1}$$

where t represents the statistical value, r represents the PCC value, and N represents the number of samples in the dataset. Accordingly, the p-value for the significance level can be derived.

In this study, the linearity between any two features in the dataset is first tested using the PCC, and if the degree of linearity of the two variables is too high, one of the variables will be discarded in the subsequent regression analysis. And then the model for predicting the contents of oxygenated components in bio-oil is trained and tested using the selected input features.

2.2. Model construction and hyperparameter optimization

The XGB algorithm is an integrated learning method used to deal with regression problems. The model input features are filtered according to the PCC. Eighty percent of the dataset is selected as the training dataset randomly, and the remaining data is used for testing. When training the prediction model using XGB, the choice of hyperparameters has a significant impact on the model results. When using the BOA for hyperparameter optimization, it is sufficient to define the optimized objective function and use the hyperparameters as inputs to the objective function. In order to obtain better model training results, the Bayesian optimization algorithm (BOA) is used to perform hyperparameter search with the regression coefficient of the validation dataset as the objective function. At the same time, the maximum number of optimizations is set to 10,000, and the optimization process is stopped when the regression coefficient of the validation dataset is stable or the maximum number of optimizations is reached. The result of the optimization is the combination of hyperparameters that maximizes the regression coefficient of the validation dataset. To avoid overfitting and ensure the generalization of the model, a 7-fold cross-validation is performed when performing hyperparameter optimization. First, the entire sample is divided into seven equally-sized subsets of samples. Then these subsets are traversed, with the current subset as the validation dataset and the rest of the samples as the training set, for training and evaluation of the model. Finally, the average of the 7-times evaluation parameters is taken as the final result.

In this model, estimators, depth, and learning rate are chosen as hyperparameters for optimization in the ranges [1, 500], [1, 50], and [0.001, 0.2], respectively. The remaining parameters of the XGB algorithm use the default values in "XGBRegressor" (gamma = 0, min_child_weight = 1, subsample = 1, colsample_bytree = 1, reg_alpha = 0, reg_lambda = 1, random_state = 0). The prediction models are constructed using the XGB algorithm through the training dataset. The model prediction performance is evaluated and compared using the test dataset. The evaluation parameters of the prediction model performance are RMSE and R^2. RMSE denotes root mean square error; R^2 denotes the regression coefficient.

In order to identify the contribution of different characteristic information of biomass to the prediction accuracy, two types of prediction models were constructed by selecting different input features: (1) A kind of prediction model (Ultimate-Component model) was constructed with biomass ultimate analysis (C-H-O-N) and pyrolysis conditions (PT-PS-FR). (2) Another kind of prediction model (Proximate-Component model) was constructed with biomass proximate analysis (FC-VM-Ash) and pyrolysis conditions (PT-PS-FR). After completing the training of the two prediction models, the model with better accuracy and generalization was selected to investigate the influence pattern of each input feature on the contents of oxygenated components using the PDA method. This method can visually represent the relationship between the features and the objective function, such as linear, monotonic, or more complex connections.

Table 1
The range of parameters of the prediction model.

Parameter Name	Parameter Range	Parameter Name	Parameter Range
Volatile matter content	42.68%–88.85%	Oxygen content	17.49%–58.73%
Ash content	0.14%–37.7%	Nitrogen content	0%–6.54%
Fixed carbon content	10.22%–35.79%	Pyrolysis temperature	150–900 °C
Carbon content	36.58%–68.19%	Particle size	150–2000 μm
Hydrogen content	4.19%–14.14%	Flow rate	0–1000 sccm

3. Results and discussion

3.1. Statistical and correlation analysis of dataset parameters

The dataset includes a wide range of biomass, like forestry waste, agricultural waste, and wood. Additionally, the dataset contains pretreated biomass, for which pretreatment methods include acid treatment, alkali treatment, baking treatment, hydrothermal treatment, etc., ensuring the generalizability of the prediction model. The range of each parameter is shown in Table 1. These input and output parameter distributions are in the appropriate ranges for the prediction model using the XGB method.

The PCCs between any two variables (Fig. 1) show that most variables are nonlinearly related to each other and are suitable as input features for constructing prediction models. If PCC is between 0.5 and 0.8, it means that the two variables are correlated; if PCC is greater than 0.8, it means that there is a very strong correlation between the two variables. A significant negative linear correlation between FC and VM (PCC greater than 0.8 and significant correlation at the $p < 0.01$ level) is found in the proximate analysis, primarily due to the fact that fixed carbon content is typically calculated using differential subtraction in the proximate analysis (FC = 100 – VM – Ash). Therefore, when training the proximate-component model, the input feature FC is disregarded. In the ultimate analysis, a negative linear correlation is also shown between C and O contents. This is because in the ultimate analysis, the O content is also obtained by differential subtraction (O = 100 – C – H – N). However, the PCC between the C and O is less than 0.8, and a previous study has shown that O content affects the contents of oxygenated components intensively (Yang et al., 2022). Therefore, both C and O contents are used as input features in the training of the Ultimate-Component model.

3.2. Model training and prediction performance

To ensure that the trained model has excellent prediction performance, the BOA is used to optimize the combination of hyperparameters during the model training process. Fig. 2a shows the change process of the validation dataset R^2 as the number of BOA optimization increases when training the Ultimate-acids model. As the number of optimizations increases, the validation dataset R^2 gradually increases. After 50 iterations of BOA optimization, the validation dataset R^2 basically does not increase anymore and remains at a high level, indicating that the optimal hyperparameter combination has been obtained. It is clear that the BOA is suitable for the hyperparameter search of the XGB algorithm. The number of estimators, the learning rate, and the max depth are gradually evolving during the optimization process. As the time for optimization increases, all three hyperparameters stabilize in a certain range. Within that range, the performance of the trained model is acceptable. The high number of estimators (Fig. 2b), the medium

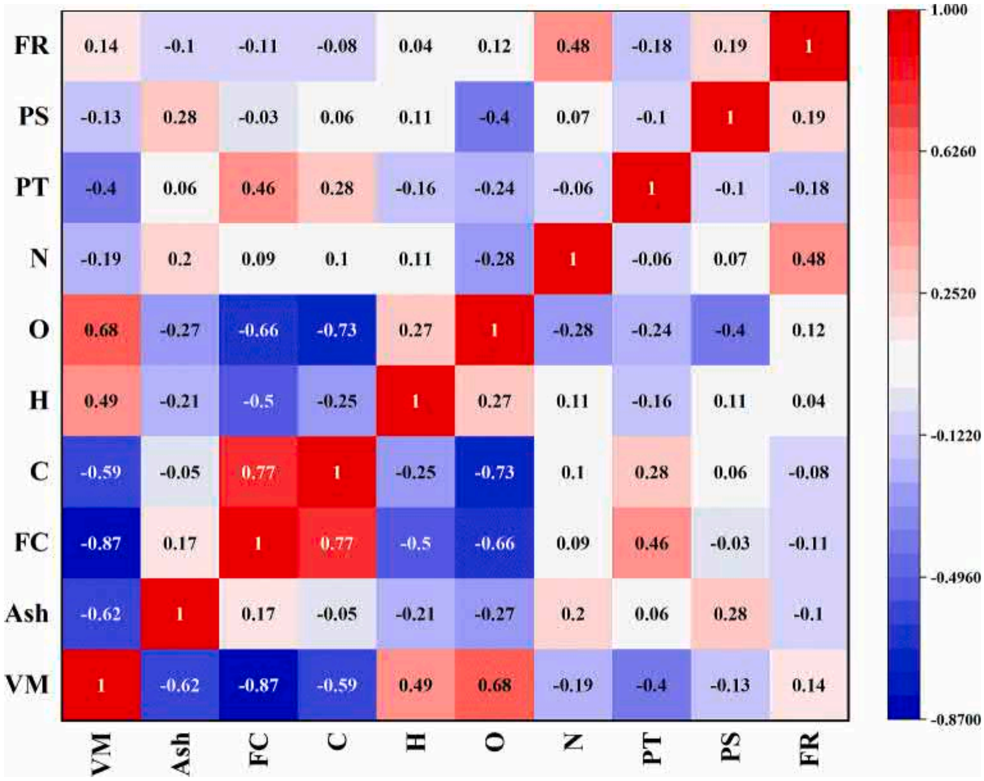


Fig. 1. Pearson correlation coefficients between any two variables.

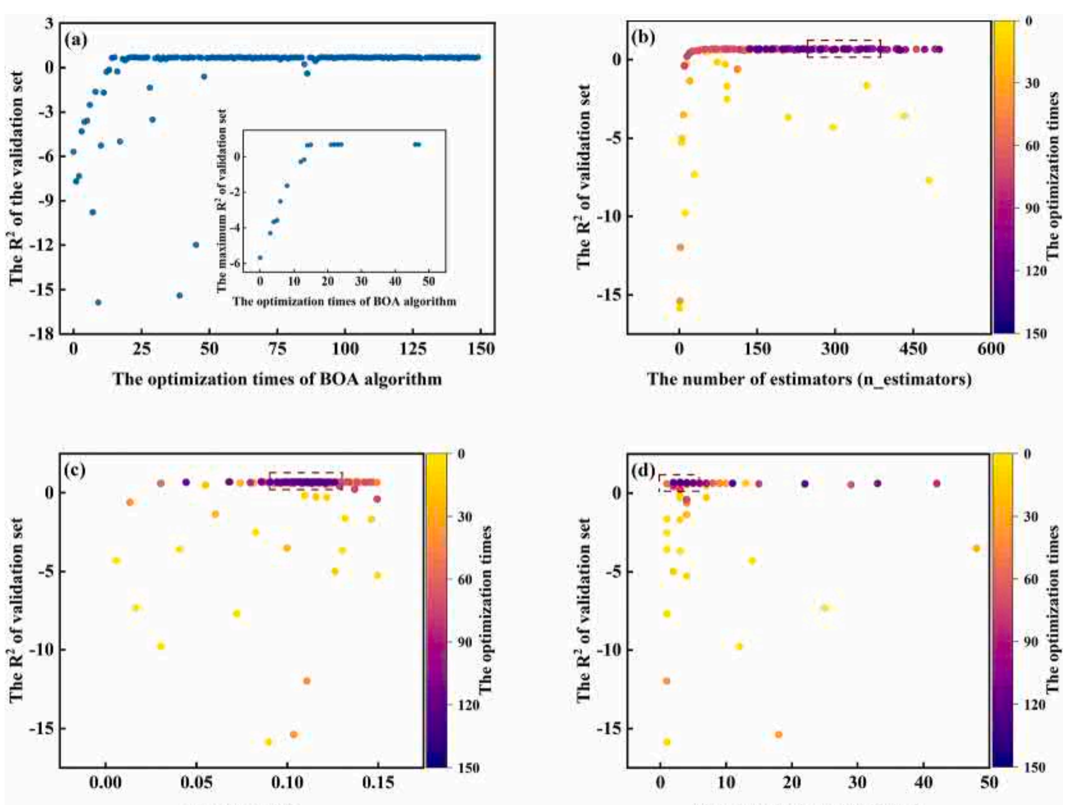


Fig. 2. Hyperparameter evolution in the optimization process of the BOA: (a) The R2 of the validation set; (b) The number of estimators; (c) Learning rate; (d)The max depth.

Table 2
Optimal hyperparameter combinations and R^2, RMSE of test dataset for each model.

Model	R^2	RMSE	Hyperparameters		
			Estimators	Learning rate	Depth
Ulimate-acids	0.9114	4.0010	318	0.1431	2
Proximate-acids	0.8231	3.9103	373	0.0823	5
Ulimate-phenols	0.8245	6.7554	436	0.1262	3
Proximate- phenols	0.7268	7.0898	149	0.0484	2
Ultimate-ketones + aldehydes	0.8557	4.2246	375	0.1953	10
Proximate-ketones + aldehydes	0.8372	4.9817	70	0.0893	5

learning rate (Fig. 2c), and the low max depth (Fig. 2d) are conducive to the training of high performance prediction models.

Twenty percent of the dataset is randomly selected as the test dataset, and the prediction model performance is evaluated using the R^2 and RMSE of the test dataset. The optimal R^2 and RMSE and the corresponding hyperparameter combinations after hyperparameter optimization for the six models are shown in Table 2.

As shown in Fig. 3, the experimental data and the output from all XGB models are compared to evaluate the model performance more intuitively. The x-axis denotes the experimental value for the content of each component class in bio-oil, and the y-axis means the predicted value for the content of each component class, where the blue line indicates the ideal situation that the predicted output is in perfect agreement with the experimental data. The closer the data points in the graph are to the blue line, the better the prediction of the model is. It can be found that the Ultimate-Component model has a higher R^2 than the Proximate-Component model when predicting the contents of oxygenated components. In addition, the Ultimate-Component model gets a smaller RMSE when predicting the contents of phenols, and the total content of ketones and aldehydes. For predicting the content of acids, the difference between the RMSEs of the two models is less, and the RMSE of the Proximate-Component model is smaller. In summary, the Ultimate-Component model is more suitable for predicting the contents of oxygenated components. In addition, the R^2 of almost all prediction models after training is greater than 0.8, and it can be considered that all models achieve good prediction results. It indicates that the XGB algorithm can predict the contents of oxygenated components in bio-oil accurately and that the BOA optimization is applicable to find the optimal combination of hyperparameters for these models.

3.3. Model input feature importance and sensitivity analysis

To analyze the extent of influence of each input feature on the contents of oxygenated components, the three best trained models

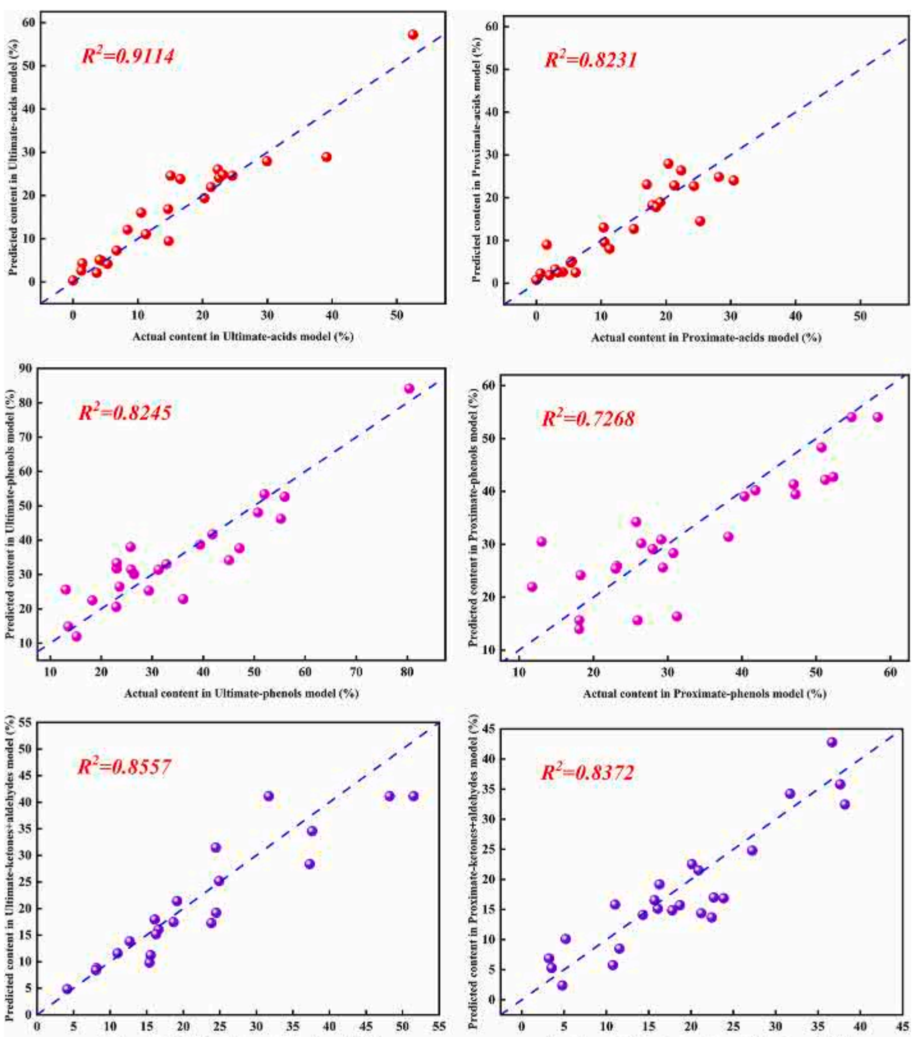


Fig. 3. Comparison of experimental data and predicted data for all XGB models: (a) Ultimate-acids; (b) Proximate-acids; (c) Ultimate-phenols; (d) Proximate-phenols; (e) Ultimate-ketones + aldehydes; (f) Proximate-ketones + aldehydes.

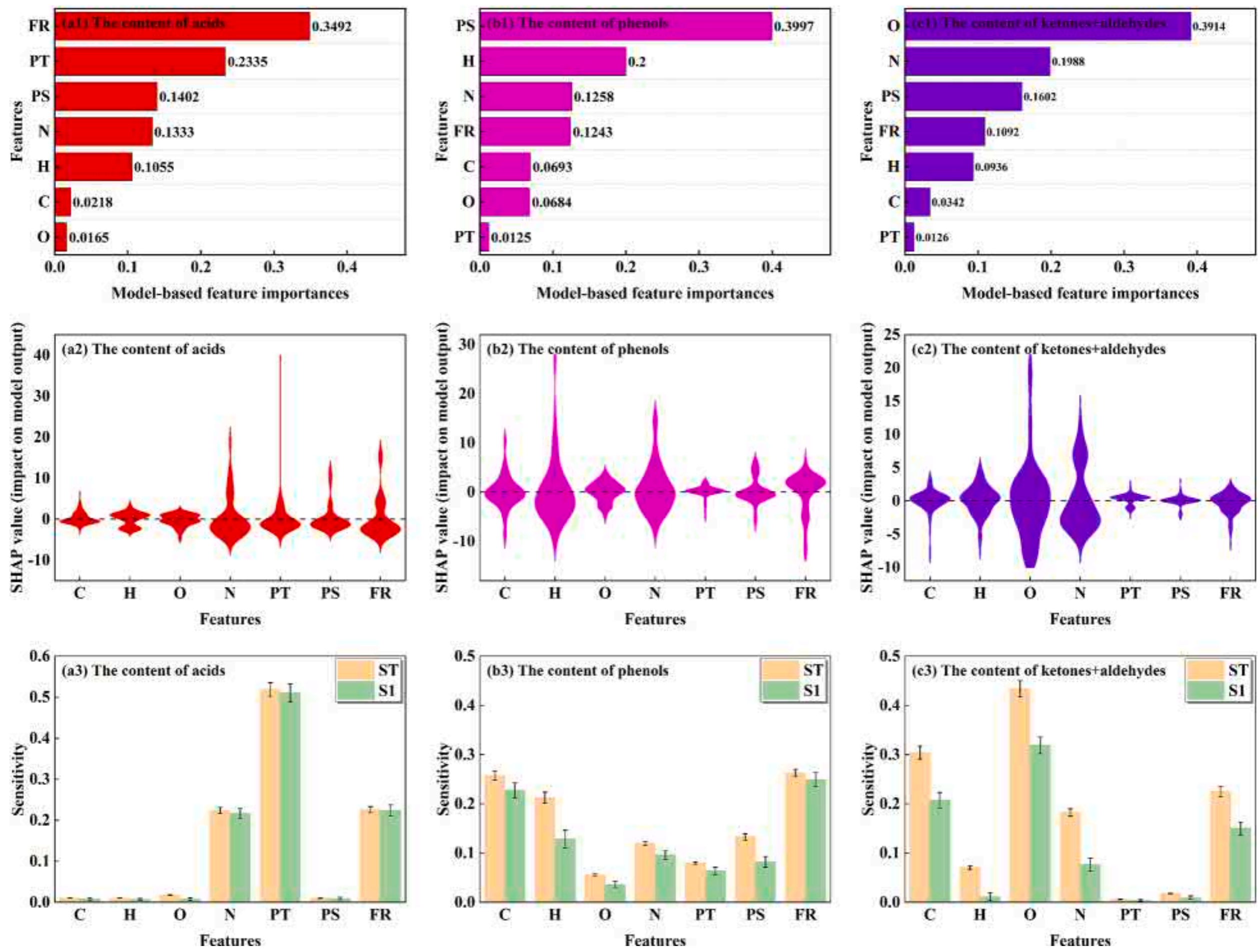


Fig. 4. Feature importance analyses for (a1) the content of acids, (b1) the content of phenols and (c1) the total content of ketones and aldehydes. SHAP value analyses for (a2) the content of acids, (b2) the content of phenols and (c2) the total content of ketones and aldehydes. And sensitivity analyses for (a3) the content of acids, (b3) the content of phenols and (c3) the total content of ketones and aldehydes.

(Ultimate-acids model, Ultimate-phenols model, and Ultimate-ketones + aldehydes model) are analyzed for the importance of input features. The methods are "xgb.feature_importances_" in the xgboost Python library and the SHAP (Shapley) value method. "xgb.feature_importances_" is a model-based feature importance analysis method that responds to the non-linear connection between each input and output variable compared to the PCC. The model-based feature importance analysis for the prediction of the content of acids is given in Fig. 4a1. The two features that contributed the most are FR and PT. This is mainly related to whether the volatile matter undergoes a secondary reaction or not. In the experiment, it is also found that the pyrolysis temperature and residence time have significant effects on the content of acids in bio-oil (Sun et al., 2016a). In Fig. 4b1, the PS has a significant impact on the prediction of the content of phenols, which might be caused by the poor distribution of biomass particle size data in the dataset. Besides, the H content in biomass is the most important feature affecting the content of phenols. The O content in biomass plays a decisive role in predicting the total content of ketones and aldehydes (Fig. 4c1).

The SHAP value method is model-independent which estimates the contribution of each input variable to the model output using the Shapley value from game theory. The central concept of the SHAP method is an additive model with all input features acting as "contributors". The model generates a prediction value for each prediction sample, and the value assigned to each feature is the SHAP value in that sample. The magnitude, positive and negative of SHAP values indicate the degree of contribution and the direction of influence of the input features on the prediction results, respectively. The width of the graph indicates the degree of concentration of the SHAP values. From Fig. 4a2, it can be seen that the effects of PT and FR are large and play a negative role in most of the samples when predicting the content of acids. The increase in pyrolysis temperature intensifies the secondary reaction of the volatile fraction and the heat transfer during the pyrolysis process, which accelerates the decomposition of oxygen-containing functional groups such as hydroxyl and carboxyl groups into small molecules (Ma et al., 2015), resulting in a decrease in acid content. FR also influences the occurrence of secondary reactions by changing the residence time (Somerville and Deev, 2020). The H content in biomass influences the content of acids positively and strongly. In Fig. 4b2, the elemental contents in biomass, especially O and H contents, exhibit large effects on the prediction of the content of phenols. In contrast, the contribution of reaction conditions to the content of phenols is not significant. The H content in biomass has a large contribution in predicting the total content of ketones and aldehydes, as shown in Fig. 4c2. This is different from the results of the model-based feature importance analysis, and the reason for this discrepancy may be the insufficient amount of data in the test dataset.

Input feature sensitivity analysis is further performed using the SALib library to quantify the extent to which changes in input features influence the contents of oxygenated components and the interactions among the input features. In Fig. 4, S1 indicates the first-order sensitivity index, representing the effect of individual input feature on the prediction result, and ST means the total-order sensitivity index indicating how the input features impact the predicted result including the

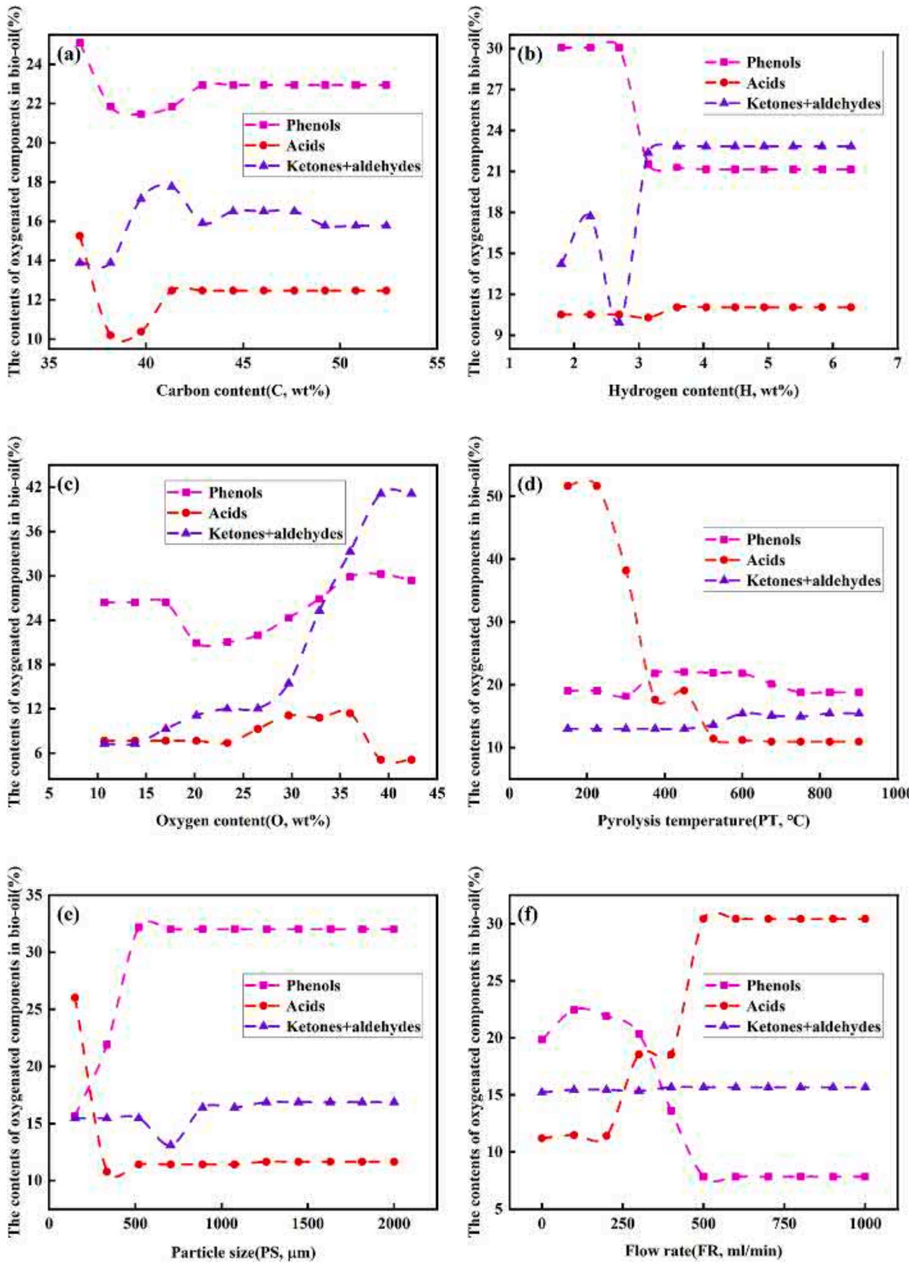


Fig. 5. One-dimensional PDA for the content of each type of oxygenated components in bio-oil versus: (a) Carbon content; (b) Hydrogen content; (c) Oxygen content; (d) Pyrolysis temperature; (e) Particle size; (f) Flow rate.

first order and higher order. The magnitude of the difference between S1 and ST can be used to determine whether there is a strong interaction between this input feature and other features. From Fig. 4a3, it is observed that the most influential input feature in predicting the content of acids is PT. Moreover, the FR (Fig. 4b3) and O content in biomass (Fig. 4c3) have the greatest effects on the content of phenols, and the total content of ketones and aldehydes. Overall, this is essentially the same as the results of the feature importance analysis. The S1 and ST for the input feature of O content in biomass in each prediction model differ significantly. This indicates a strong interaction between O content in biomass and other features, which will be discussed in the subsequent two-dimensional PDA.

3.4. One-dimensional PDA for the contents of oxygenated components

The PDA is generally used to investigate the change pattern of the target variable (the content of each component class in the bio-oil) with the input features by adjusting specific input features while averaging the remaining input features. Fig. 5 depicts a one-dimensional PDA plot for the content of each type of oxygenated components. However, the result from the PDA analysis of nitrogen content in biomass is not shown because the nitrogen content is generally low and of low impact.

A higher oxygen content in biomass can result in a higher oxygen content because it is generally difficult to remove oxygen from biomass by decarboxylation or decarbonylation in the absence of the catalyst (Wu et al., 2022). At the low oxygen content in biomass (<15%), the oxygenated component class is mainly phenols. When the oxygen content in biomass is slightly higher (15% – 35%), the content of phenols in bio-oil decreases, and the total content of ketones and aldehydes and the content of acids increase slightly. If the oxygen content in biomass continues to increase, the total content of ketones and aldehydes increases rapidly, while the content of acids decreases, and the content of phenols increases slightly and then remains stable. This is mainly determined by the redox relationship between the hydroxyl, aldehydes,

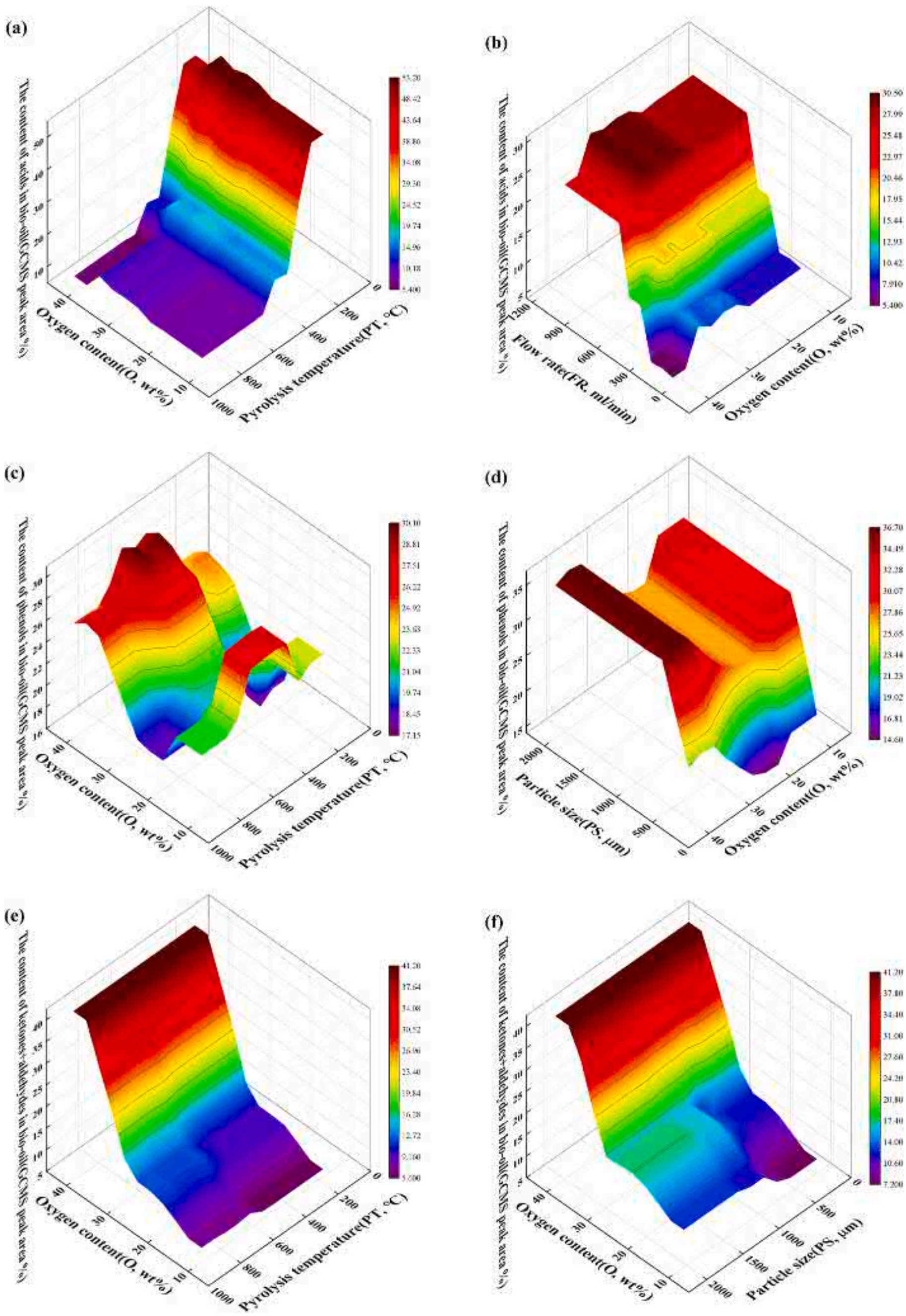


Fig. 6. Two-dimensional PDA for the contents of oxygenated components in bio-oil: (a) The content of acids–O and PT; (b) The content of acids–O and FR; (c) The content of phenols–O and PT; (d) The content of phenols–O and PS; (e) The total content of ketones and aldehydes–O and PT; (f) The total content of ketones and aldehydes–O and PS.

and carboxyl groups. At the low carbon and hydrogen contents in biomass, their effects on the contents of oxygenated products are more pronounced. However, when the carbon content and hydrogen content in biomass are above a certain standard (C: 45%, H: 3.5%), the content of each oxygenated component class in bio-oil remains essentially stable. This indicates that the key factor affecting the contents of oxygenated components is the oxygen content in biomass.

When the PT is less than 400 °C, bio-oil contains more acids, which can account for more than 50%. As the temperature increases, the content of acids gradually decreases. This is because the decomposition temperature of hemicellulose is generally 200 – 315 °C, and the acids are mainly from hemicellulose (Wang et al., 2013b). As the PT increases, C – O, C – H, and C – C break in the order of weak to strong bond energy and undergo decarboxylation, decarbonylation, dehydrogenation, and aromatization reactions (Serio et al., 1987), and acids are converted into more stable phenols and aromatic compounds. This is one of the reasons why the content of phenols increases after 400 °C. Another reason for this increase is that the cellulose is almost completely decomposed, and the lignin in the biomass also starts to decompose at this time. However, continuing to increase the temperature to 600 °C will cause the phenols to decompose (Yang et al., 2020). In addition, the increase of the total content of ketones and aldehydes mainly occurs at high temperatures, because lignin decomposes more completely at higher temperatures to produce more ketones and aldehydes.

The influence of PS on the contents of bio-oil components occurs only in a small range (Shen et al., 2009). Meanwhile, an insufficient amount of data with PS larger than 1000 may also lead to such a result. With the increase of the PS, the content of acids in bio-oil decreases and the content of phenols increases when the PS is less than 500 μm, while the total content of ketones and aldehydes remains basically stable. This is because biomass particle size affects biomass pyrolysis behavior by changing heat and mass transfer. Ketones and aldehydes in bio-oil are mainly produced at high pyrolysis temperatures, when the influence of heat transfer processes is not sufficiently pronounced. The FR in biomass pyrolysis process affects the residence time of reactants and the intensity of secondary reactions. The content of acids and the FR generally show a positive correlation, while the content of phenols and the FR show a negative correlation. In addition, the impact of the FR on the total content of ketones and aldehydes is not obvious.

3.5. Two-dimensional PDA for the contents of oxygenated components

Two-dimensional PDA was used to explore how any two features together affect the contents of oxygenated components. The interaction between the biomass characteristics and the reaction conditions is represented visually. Some typical two-dimensional PDA plots for the contents of oxygenated components are given in Fig. 6. The above one-dimensional PDA analysis elucidates the importance of oxygen content in biomass for the contents of oxygenated components. Therefore, the analysis for two-dimensional PDA will focus on the interaction between oxygen content in biomass and pyrolysis conditions. Because high temperature accelerates the heat transfer during pyrolysis, it can effectively promote the decomposition reaction of oxygen-containing functional groups (carboxyl groups, et al.) into small molecule products. Thus, higher oxygen content in biomass and higher pyrolysis temperature can extremely reduce the content of acids. At low flow rates, the change in the content of acids is more pronounced with the change in oxygen content in biomass because the low flow rate contributes to the occurrence of volatile fraction secondary reactions, while the production of acids is more dependent on the volatile fraction secondary reactions.

The decomposition of lignin is the main pathway for producing phenols. When the PT is high (400 – 600 °C), the lignin in biomass is basically able to decompose completely. And with the increase of oxygen content in biomass providing more reactants for the reaction, the content of phenols rises significantly. However, if the pyrolysis temperature continues to increase (>600 °C), the decomposition of some phenols occurs at this temperature range, and the increased oxygen content in biomass does not have large effect on the content of phenols in bio-oil any more. This effect is more pronounced when the PS is small, because small PS contributes to heat and mass transfer during pyrolysis of biomass, and the lignin can be more completely decomposed. When the oxygen content in biomass is high, the hydroxyl groups in the pyrolysis products are more easily oxidized to aldehydes groups. Thus, the total content of ketones and aldehydes remains at a high yield as the oxygen content of biomass rises, and the effects of PT and PS on this content decreases.

In summary, the two-dimensional PDA for the contents of oxygenated components in bio-oil clearly shows the interaction between biomass properties and pyrolysis conditions. This not only contributes to the understanding of reaction mechanism of biomass pyrolysis but also facilitates the optimization of biomass pretreatment process. By adjusting pretreatment process to improve biomass properties while targeting pyrolysis conditions, higher quality of bio-oil can be obtained, and even targeted regulation of pyrolysis products from biomass can be achieved.

4. Conclusions

The models based on the XGB algorithm can accurately predict the contents of oxygenated components in bio-oil. The models constructed using biomass ultimate analysis and pyrolysis conditions have better prediction accuracy and generalization. Understanding the composition of bio-oil no longer relies on complex experiments, which helps rapidly optimize biomass pretreatment technologies and pyrolysis processes. However, machine learning is a data-driven approach that presents challenges in interpreting reaction mechanisms. In future research, combining machine learning models with physical information, such as reaction kinetics, thermodynamics, and transfer mechanisms, will further improve the performance of machine learning models and result in deeper insights.

Data availability

The data can be found at https://github.com/Buaa-su/XGB.git.

CRediT authorship contribution statement

Sheng Su: Investigation, Software, Methodology, Validation, Data curation, Writing – original draft. **Juan Wang:** Writing – review & editing, Supervision, Project administration, Funding acquisition.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgements

This research was supported by the National Natural Science Foundation of China under grant numbers U2032119 and 71690245.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.biortech.2023.129040.

References

Bi, D., Jiang, M., Lin, X., Huang, F., Liu, S., 2020. Effects of carbamide pretreatment on product distribution and bio-oil composition from fast pyrolysis of pine powder. Biomass Convers. Biorefin. 12, 3593–3602.

Cen, K., Zhang, J., Ma, Z., Chen, D., Zhou, J., Ma, H., 2019. Investigation of the relevance between biomass pyrolysis polygeneration and washing pretreatment under different severities: Water, dilute acid solution and aqueous phase bio-oil. Bioresour. Technol. 278, 26–33.

Chen T., He T., Benesty M., Khotilovich V., 2015b. Xgboost: extreme gradient boosting. R package version 0.4.2 1(4), 1-4.

Chen T., Guestrin C., Assoc Comp M., 2016. XGBoost: A Scalable Tree Boosting System. 22nd ACM SIGKDD. San Francisco, CA. pp. 785-794.

Chen, D., Zheng, Z., Fu, K., Zeng, Z., Wang, J., Lu, M., 2015. Torrefaction of biomass stalk and its effect on the yield and quality of pyrolysis products. Fuel 159, 27–32.

Chen, D., Cen, K., Jing, X., Gao, J., Li, C., Ma, Z., 2017. An approach for upgrading biomass and pyrolysis product quality using a combination of aqueous phase bio-oil washing and torrefaction pretreatment. Bioresour. Technol. 233, 150–158.

Chen, D., Cen, K., Cao, X., Zhang, J., Chen, F., Zhou, J., 2020. Upgrading of bio-oil via solar pyrolysis of the biomass pretreated with aqueous phase bio-oil washing, solar drying, and solar torrefaction. Bioresour. Technol. 305, 123130.

Chen, D., Cen, K., Cao, X., Chen, F., Zhang, J., Zhou, J., 2021. Insight into a new phenolic-leaching pretreatment on bamboo pyrolysis: Release characteristics of pyrolytic volatiles, upgradation of three phase products, migration of elements, and energy yield. Renew. Sustain. Energy Rev. 136, 110444.

Chen, X., Zhang, H., Song, Y., Xiao, R., 2018. Prediction of product distribution and bio-oil heating value of biomass fast pyrolysis. Chem. Eng. Process. 130, 36–42.

Dhyani, V., Bhaskar, T., 2018. A comprehensive review on the pyrolysis of lignocellulosic biomass. Renew. Energy 129, 695–716.

Dong, Z., Bai, X., Xu, D., Li, W., 2023. Machine learning prediction of pyrolytic products of lignocellulosic biomass based on physicochemical characteristics and pyrolysis conditions. Bioresour. Technol. 367, 128182.

Ge, S., Shi, Y., Xia, C., Huang, Z., Manzo, M., Cai, L., Ma, H., Zhang, S., Jiang, J., Sonne, C., Lam, S.S., 2021. Progress in pyrolysis conversion of waste into value-added liquid pyro-oil, with focus on heating source and machine learning analysis. Energy Convers. Manage. 245, 114638.

International Energy Agency, 2021. World energy outlook 2021.

Lu, X., Jiang, J., He, J., Sun, K., Sun, Y., 2020. Synergy of Hydrothermal and Organic Acid Washing Treatments in Chinese Fir Wood Vinegar Preparation. ACS Omega 5 (23), 13685–13693.

Lu, X., Han, T., Jiang, J., Sun, K., Sun, Y., Yang, W., 2020. Comprehensive insights into the influences of acid-base properties of chemical pretreatment reagents on biomass pyrolysis behavior and wood vinegar properties. J. Anal. Appl. Pyrolysis 151, 104907.

Ma, Z., Chen, D., Gu, J., Bao, B., Zhang, Q., 2015. Determination of pyrolysis characteristics and kinetics of palm kernel shell using TGA–FTIR and model-free integral methods. Energy Convers. Manage. 89, 251–259.

Mathur, J., Baruah, B., Tiwari, P., 2022. Prediction of bio-oil yield during pyrolysis of lignocellulosic biomass using machine learning algorithms. Can. J. Chem. 101 (5), 2457–2471.

Nielsen, D., 2016. Tree boosting with xgboost-why does xgboost win“ every” machine learning competition. NTNU, Trondheim.

Pathy, A., Meher, S., Balasubramanian, P., 2020. Predicting algal biochar yield using eXtreme Gradient Boosting (XGB) algorithm of machine learning methods. Algal Res. 50, 102006.

Qiu, B., Tao, X., Wang, J., Liu, Y., Li, S., Chu, H., 2022. Research progress in the preparation of high-quality liquid fuels and chemicals by catalytic pyrolysis of biomass: A review. Energy Convers. Manage. 261, 115647.

Rezaei, P.S., Shafaghat, H., Daud, W.M.A.W., 2014. Production of green aromatics and olefins by catalytic cracking of oxygenate compounds derived from biomass pyrolysis: A review. Appl. Catal A 469, 490–511.

Serio, M.A., Hamblen, D.G., Markham, J.R., Solomon, P.R., 1987. Kinetics of volatile product evolution in coal pyrolysis: experiment and theory. Energy Fuels 1 (2), 138–152.

Setkit, N., Li, X., Yao, H., Worasuwannarak, N., 2021. Torrefaction under mechanical pressure of 10–70 MPa at 250 °C and its effect on pyrolysis behaviours of leucaena wood. Bioresour. Technol. 338, 125503.

Shen, J., Wang, X., Garcia-Perez, M., Mourant, D., Rhodes, M.J., Li, C., 2009. Effects of particle size on the fast pyrolysis of oil mallee woody biomass. Fuel 88 (10), 1810–1817.

Somerville, M., Deev, A., 2020. The effect of heating rate, particle size and gas flow on the yield of charcoal during the pyrolysis of radiata pine wood. Renew. Energy 151, 419–425.

Sun, J., He, F., Pan, Y., Zhang, Z., 2016. Effects of pyrolysis temperature and residence time on physicochemical properties of different biochar types. Acta Agric. Scand., Sect. B 67 (1), 12–22.

Sun, Y., Liu, L., Wang, Q., Yang, X., Tu, X., 2016. Pyrolysis products from industrial waste biomass based on a neural network model. J. Anal. Appl. Pyrolysis 120, 94–102.

Sun, Y., He, Z., Tu, R., Wu, Y., Jiang, E., Xu, X., 2019. The mechanism of wet/dry torrefaction pretreatment on the pyrolysis performance of tobacco stalk. Bioresour. Technol. 286, 121390.

Tang, Q., Chen, Y., Yang, H., Liu, M., Xiao, H., Wu, Z., Chen, H., Naqvi, S.R., 2020. Prediction of bio-oil yield and hydrogen contents based on machine learning method: effect of biomass compositions and pyrolysis conditions. Energy Fuels 34 (9), 11050–11060.

Tripathi, M., Sahu, J.N., Ganesan, P., 2016. Effect of process parameters on production of biochar from biomass waste through pyrolysis: A review. Renew. Sustain. Energy Rev. 55, 467–481.

Tu, R., Sun, Y., Wu, Y., Fan, X., Cheng, S., Jiang, E., Xu, X., 2021. Selective production of furfural and phenols from rice husk: the influence of synergetic pretreatments with different order. Renew. Energ. 168, 297–308.

Vuppaladadiyam, A.K., Vuppaladadiyam, S.S.V., Awasthi, A., Sahoo, A., Rehman, S., Pant, K.K., Murugavelh, S., Huang, Q., Anthony, E., Fennel, P., Bhattacharya, S., Leu, S., 2022. Biomass pyrolysis: A review on recent advancements and green hydrogen production. Bioresour. Technol. 364, 128087.

Wang, H., Male, J., Wang, Y., 2013. Recent advances in hydrotreating of pyrolysis bio-oil and its oxygen-containing model compounds. ACS Catal. 3 (5), 1047–1070.

Wang, S., Ru, B., Lin, H., Luo, Z., 2013. Degradation mechanism of monosaccharides and xylan under pyrolytic conditions with theoretic modeling on the energy profiles. Bioresour. Technol. 143, 378–383.

Wu, K., Luo, B., Wu, H., Chen, M., Wang, Y., Ge, Z., Chu, S., Liu, Q., Zhang, H., 2022. Enhancement of the production of chemicals and liquid fuels from grass biowaste via NaOH-Fenton pretreatment coupled with fast pyrolysis. Energy Convers. Manage. 251, 114954.

Xu, B., Chen, T.C., Ahangari, D., Alizadeh, S.M., Elveny, M., Makhdoumi, J., 2021. Application of a supervised learning machine for accurate prognostication of hydrogen contents of bio-oil. Int. J. Chem. Eng. 2021, 7548251.

Yang, K., Wang, J., Huang, J., Yang, J., Pan, Y., Xu, M., Ma, P., Jia, L., 2020. Understanding the homogeneous reactions of primary tar from biomass pyrolysis by means of photoionization mass spectrometry. Energy Fuels 34 (10), 12678–12687.

Yang, K., Wu, K., Zhang, H., 2022. Machine learning prediction of the yield and oxygen content of bio-oil via biomass characteristics and pyrolysis conditions. Energy 254, 124320.

Zeng, K., He, X., Yang, H., Wang, X., Chen, H., 2019. The effect of combined pretreatments on the pyrolysis of corn stalk. Bioresour. Technol. 281, 309–317.

Zhang, T., Cao, D., Feng, X., Zhu, J., Lu, X., Mu, L., Qian, H., 2022. Machine learning prediction of bio-oil characteristics quantitatively relating to biomass compositions and pyrolysis conditions. Fuel 312, 122812.

Zhang, S., Dong, Q., Chen, T., Xiong, Y., 2016. Combination of light bio-oil washing and torrefaction pretreatment of rice husk: its effects on physicochemical characteristics and fast pyrolysis behavior. Energy Fuels 30 (4), 3030–3037.

Zhang, Y., Lv, P., Wang, J., Wei, J., Cao, P., Bie, N., Bai, Y., Yu, G., 2022. Product characteristics of rice straw pyrolysis at different temperature: Role of inherent alkali and alkaline earth metals with different occurrence forms. J. Energy Inst. 101, 201–208.

Zhang, S., Xiong, Y., 2016. Washing pretreatment with light bio-oil and its effect on pyrolysis products of bio-oil and biochar. RSC Adv. 6 (7), 5270–5277.

Zhang, S., Zhu, S., Zhang, H., Liu, X., Zhang, H., 2019. Evaluation of pyrolysis behavior and products properties of rice husk after combined pretreatment of washing and torrefaction. Biomass Bioenergy 127, 105293.

Zhuang, X., Gan, Z., Chen, D., Cen, K., Ba, Y., Jia, D., 2022. An approach for upgrading bio-oil by using heavy bio-oil co-pyrolyzed with bamboo leached with light bio-oil. Fuel 331 (2), 125931.